# Oriol Vallès Codina

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EU National | US Citizen

## RESEARCH AND TEACHING INTERESTS

- Macroeconomics (E3)
- Economic Development, Innovation, Technological Change, and Growth (O25, O41, O44)
- Environmental and Ecological Economics (Q56, Q57, Q32)
- International Economics (F02, F4)
- Mathematical and Quantitative Methods, Econometrics (C55, C61, C63, C67, D85)
- History of Economic Thought: Classical Political Economy (B12, B51)

## EDUCATION

**Ph.D. in Economics** 2014–2021
*New School for Social Research* New York, NY, US
Dissertation: "Economic Production as Life: A Classical Approach to Computational Social Science"
Advisors: Anwar Shaikh, Duncan Foley, Willi Semmler

**M.A. in Anthropology** 2011–2014
*Columbia University of New York* New York, NY, US
Thesis: "Political Economy of the Spanish Housing Bubble, 1998-2008"
Readers: Michael Blim, Terence D'Altroy

**M.Sc. in Bioinformatics** 2009–2011
*Universitat Pompeu Fabra* Barcelona, ES

**B.Sc. in Physics** 2003–2009
*Universitat Autònoma de Barcelona* Barcelona, ES

## WORK EXPERIENCE

**Senior Research Associate** 2025–Present
*Net Zero Industrial Policy Lab, Johns Hopkins University* Baltimore, US
Leads the empirical mapping of global clean-technology value chains (batteries, solar, wind, electrolyzers, heat pumps, nuclear, transmission) from bilateral trade and multi-region input-output data; builds and maintains the lab's trade-based analytic platform for policy audiences.

**Research Consultant** 2024–2025
*World Bank* Washington DC, US
*Agence Française de Développement* Paris, FR
*De la Salle University* Manila, PH
Developed ecologically-extended macroeconomic models for Mexico and the Philippines.

**Postdoctoral Research Fellow** 2022–2024
*Leeds University Business School* Leeds, UK

**Postdoctoral Research Fellow in Innovation Policy** 2020–2021
*Institute for Innovation and Public Purpose, University College London* London, UK

**Teaching Assistant and Fellow, Research Fellow** 2016–2020
*New School for Social Research* New York, NY, US

## PUBLICATIONS

### Peer-Reviewed Articles

- **Vallès Codina, O.**, Fevereiro, J. B. R. T., Genovese, A., Kaltenbrunner, A., Kesidou, E. & Passarella, M. V. (2026). "The Macroeconomic Consequences of Ecological Unequal Exchange" *Ecological Economics* (revise and resubmit).

- Fevereiro, J. B. R. T., Genovese, A., Purvis, B., Passarella, M. V. & **Vallès Codina, O.** (2025). "Macroeconomic Models for Assessing the Transition towards a Circular Economy: A Review" *Ecological Economics*. [link]

- **Vallès Codina, O.** & Semmler, W. (2024). "Time Scales of the Low-Carbon Transition: A Data-Driven Dynamic Multi-Sector Growth Model" *Journal of Economics and Statistics (Jahrbücher für Nationalökonomie und Statistik)*. [link]

- **Vallès Codina, O.** (2023). "Business cycles, sectoral price stabilization, and climate change mitigation: A model of multi-sector growth in the tradition of the Bielefeld disequilibrium approach" *Journal of Economic Behavior & Organization*, 216, 636–653. [link]

- Hartasánchez, D. A., Brasó-Vives, M., Fuentes-Díaz, J., **Vallès Codina, O.** & Navarro, A. (2016). "SeDuS: segmental duplication simulator" *Bioinformatics*, 32(1), 148–150. [link]

- Hartasánchez, D. A., **Vallès Codina, O.**, Brasó-Vives, M. & Navarro, A. (2014). "Interplay of Interlocus Gene Conversion and Crossover in the Molecular Evolution of Segmentally Duplicated Sequences: the Neutral Scenario" *G3*, 4(8), 1479–1489. [link]

- **Vallès Codina, O.**, Möbius, R., Rüdiger, S. & Schimansky-Geier, L. (2011). "Traveling echo waves in an array of excitable elements with time-delayed coupling" *Physical Review E*, 83(3). [link]

### Books

- Booss-Bavnbek, B., Christensen, J. H., Richardson, K. & **Vallès Codina, O.** (2024). *Multiplicity of Time Scales in Complex Systems: Challenges for Sciences and Dissemination (Volumes I & II)*. Springer/Birkhauser. [link]

### Book Chapters

- Booss-Bavnbek, B., Christensen, J. H., Richardson, K. & **Vallès Codina, O.** (2024). "Introduction" In *Multiplicity of Time Scales in Complex Systems Vol. I, Springer/Birkhauser*.

- Bimpizas-Pinis, M., Genovese, A., Kaltenbrunner, A., Kesidou, E., Purvis, B., Fevereiro, J. B. R. T., **Vallès Codina, O.** & Veronese Passarella, M. (2023). "Using input-output stock-flow consistent models to simulate and assess circular economy strategies" In *Circular Economy for Social Transformation, JUST2CE E-book*. [link]

- Holloway, C., Barbareschi, G., **Vallès Codina, O.**, Colton, N., Eide, A. H., Morgado-Ramirez, Z. D., Danemayer, J., Kattel, R. & Austin, V. (2023). "A human rights-based approach to assistive technology provision in global policy" In *Research Handbook on Disability Policy, Edward Elgar*, pp. 807–822. [link]

### Encyclopedia Entries

- **Vallès Codina, O.** (2022). "Discrete-Time Models; Punctuated Equilibrium Theory; Subsistence; System Dynamic Models; Temporal Dynamics" In *Dictionary of Ecological Economics, Edward Elgar (eds. Haddad B. & Solomon B.D.)*. [link]

### Theses

- **Vallès Codina, O.** (2021). *Economic Production as Life: A Classical Approach to Computational Social Science*. PhD thesis, The New School for Social Research.
- **Vallès Codina, O.** (2014). *The Political Economy of the Spanish Housing Bubble: 1998-2008*. MA thesis, Columbia University.

### Working Papers

- **Vallès Codina, O.** (2022). "The Purely Economic Case for Investing in Health for All in Lower- and Upper-Middle Income Countries: input-output and network analysis of their industrial ecosystem" UCL IIPP Working Paper Series (IIPP WP 2022-03). [link]
- **Vallès Codina, O.** (2020). "Classical-Evolutionary Dynamics of Price Formation" NSSR Working Paper, New School for Social Research. [link]

### Work in Progress

- **Vallès Codina, O.** "Supplier of Last Resort: Public Buffer Stocks and Price Stabilization in a Multi-Sector Economy with Exhaustible Resources" For *Structural Change and Economic Dynamics*.
- Cserjan, L. & **Vallès Codina, O.** "The core-periphery structure of global trade" For *Review of International Political Economy*.
- **Vallès Codina, O.** & Chatzarakis, N. "A thermodynamic approach to classical political economy" For *Review of Political Economy*.
- **Vallès Codina, O.** & Rodousakis, N. "Cycles of migration in coupled Goodwin models".
- **Vallès Codina, O.** "A conflict theory of pricing" For *Industrial and Corporate Change*.
- **Vallès Codina, O.** "Classifying prototypes of national industrial structures".
- **Vallès Codina, O.** "Is Health at Odds with the Economy? Input-Output and Network Analysis of the Health Industrial Ecosystem in Middle-Income Economies" For *Economic Systems Research*.
- **Vallès Codina, O.** "No evidence of long-term regional income convergence in Europe 1900-2015".
- **Vallès Codina, O.** "Review of "Marx's Theory of Value at the Frontiers" by Güney Işikara and Patrick Mokre" For *Review of Political Economy*.
- **Vallès Codina, O.** "The Conservation of Value: The Fundamental Laws of Motion of Value and Social Reproduction" For *Review of Political Economy*.
- **Vallès Codina, O.** "The Sraffian Economy as an Autocatalytic Set" For *Physical Review E*.
- **Vallès Codina, O.**, Rodousakis, N., Soklis, G. & Yajima, T. G. "Input-output analysis of global value chains of German energy".
- **Vallès Codina, O.** "Environmentalism without class struggle is just gardening: the Ricardian dynamics of the metabolic rift" For *Ecological Economics*.

## PROFESSIONAL SERVICE

### Editorial Service

- **Guest Editor**, *Multiplicity of Time Scales in Complex Systems: Challenges for Sciences and Dissemination* (Volume I and II), Springer/Birkhauser 2024

### Referee Service

- *Springer/Nature Mathematics*
- *Review of Political Economy*
- *Review of Evolutionary Political Economy*
- *Jahrbücher für Nationalökonomie und Statistik*

## FELLOWSHIPS AND AWARDS

- JUST2CE – A Just Transition to the Circular Economy 2022–2024
- Growinpro - Growth Welfare Innovation Productivity 2020–2021
- New School for Social Research Dean's Merit Scholarship 2014–2016
- La Caixa Foundation Fellowship for Graduate Study 2011–2013
- FARO Global Fellowship 2009
- Erasmus Fellowship 2007–2008

## CONFERENCES AND WORKSHOP PRESENTATIONS

### Invited Talks and Workshops

- "The Macroeconomic Consequences of Ecological Unequal Exchange" 2026
  *3rd International Workshop on Empirical SFC Models, Cassino University* Cassino, IT
- "The Macroeconomic Consequences of Ecological Unequal Exchange" 2025
  *Forum for Macroeconomics and Macroeconomic Policies, Hans-Boeckler Stiftung* Berlin, DE
- "Stock-flow Consistent Macroeconomics in the Face of the Climate Crisis" 2024
  *Angelo King Institute, De la Salle University* Manila, PH
- "Meeting the Challenges of Climate Change" 2023
  *New School for Social Research* New York, NY, US
- "Towards a Macroeconomics of the Just Transition to a Circular Economy: Stabilizing Unstable Economy-Ecology Interactions" 2023
  *Levy Economics Institute at Bard College* Red Hook, NY, US
- "Stabilizing Economic Fluctuations and Economy-Environment Interactions with a Novel Macroeconomic Model of Multi-Sector Growth: Ecological Applications of the Bielefeld Disequilibrium Approach" 2022
  *Forum for Macroeconomics and Macroeconomic Policies, Hans-Boeckler Stiftung* Berlin, DE
- "Introduction to Coding Dynamical Systems and Agent-Based Models in Ecological Economics and Classical Political Economy" 2022
  *Bamberg University* Bamberg, DE
- "Classical Political Economy in the Context of the Uneven Climate Crisis" 2022
  *University of València* València, ES
- "Agent-based Classical Competition" 2021
  *First International Workshop on Agentization, George Mason University* Fairfax, VA, US
- "Multiplicity of Time Scales in Economic Systems" 2021
  *Conference on "Multiplicity of Time Scales in Complex Systems: Challenges for Sciences and Dissemination", Roskilde University* Copenhagen, DK
- "The Green Transition" 2020
  *Institute for Innovation and Public Purpose, University College London* London, UK
- "Classical-Evolutionary Dynamics of Price Formation" 2019
  *Oeconomia - History, Methodology, Philosophy, Paris 1 University* Paris, FR

- "The Perils Anew of Peripheralization for Italy and Its Regions" 2018
  *Italian Studies Seminar, Columbia University* New York, NY, US

**Panels Organized**

- "Stock-Flow Consistent Input/Output Macroeconomics" 2023
  *International Input-Output Association Conference* Alghero, IT
- "Climate Change" 2021
  *Eastern Economic Association Conference* New York, NY, US

**Conference Papers**

- "An Ecological Interpretation of the Ricardian Growth Model" 2024
  *Ecological Economics Conference* Pontevedra, ES
- "An Ecological Interpretation of the Ricardian Growth Model" 2024
  *Eastern Economic Association* Boston, MA, US
- "The Macroeconomics of a Socially Just Transition to a Circular Economy" 2024
  *Eastern Economic Association* Boston, MA, US
- "Input-Output and Network Analysis of the Industrial Ecosystem of Health and Assistive Technology Industries" 2022
  *European Association for Evolutionary Political Economy* Napoli, IT
- "Input-Output and Network Analysis of the Industrial Ecosystem of Health and Assistive Technology Industries" 2022
  *Association for Heterodox Economics* London, UK
- "Assessing the Speed of Substitution of Carbon Technologies" 2021
  *Forum for Macroeconomics and Macroeconomic Policies, Hans-Boeckler Stiftung* Berlin, DE
- "The Green Transition" 2021
  *Eastern Economic Association Conference* New York, NY, US
- "Directed Technical Change Towards Decarbonization" 2020
  *14th International Computational and Financial Econometrics Conference* London, UK
- "Evolutionary Dynamics of Real Competition" 2020
  *New School for Social Research Economics Brown Bag Seminar* New York, NY, US
- "An Evolutionary Approach to Real Competition" 2019
  *Eastern Economic Association Conference* New York, NY, US
- "Real Competition Within a Single Industry" 2018
  *History of Economics Society Conference* Chicago, IL, US
- "The Perils Anew of Peripheralization for Italy and Its Regions" 2017
  *Capitalist Development in Hostile Environments Conference* Calabria, IT

## TEACHING EXPERIENCE

- **Social Networks** 2023
  *University of Massachusetts Amherst* Amherst, MA, US
  Corresponding Instructor
- **Introduction to Coding Dynamical Systems and Agent-Based Models** 2022
  *University of Bamberg* Bamberg, DE
  Sole Instructor

- **Computational Workshop on Dynamical Systems** 2020
  *New School for Social Research* New York, NY, US
  Sole Instructor

- **Advanced Mathematical Methods for Economic Modeling** 2020
  *New School for Social Research* New York, NY, US
  Graduate Teaching Assistant

- **Historical Foundations of Political Economy** 2016
  *New School for Social Research* New York, NY, US
  Graduate Teaching Assistant

- **Introduction to Political Economy** 2016–2020
  *Eugene Lang College, The New School* New York, NY, US
  Undergraduate Teaching Assistant

- **Origins of the Crisis in the Eurozone** 2016–2019
  *Eugene Lang College, The New School* New York, NY, US
  Teaching Fellow (Sole Instructor)

## RESEARCH EXPERIENCE

- **Research Assistant to Paulo dos Santos** 2017–2018
  *New School for Social Research* New York, NY, US
  Produced agent-based simulations of competitive price-setting firms in a simple classical circuit-of-capital framework.

- **Research Assistant to William Milberg, Cédric Durand** 2015–2017
  *New School for Social Research* New York, NY, US
  Parsed data from OECD, WIOD, and Compustat databases; econometrics on intellectual property monopoly and firm growth.

- **Research Assistant to Michael Blim** 2014
  *Dept. of Anthropology, Graduate Center, CUNY* New York, NY, US
  Performed cluster analysis on the complex core-periphery structure of the Italian and European economic geography.

- **Research Assistant to Arcadi Navarro** 2010–2011
  *Barcelona Biomedical Research Park* Barcelona, ES
  Programmed computer simulations of large-scale genetic populations.

- **Research Assistant to Lutz Schimansky-Geier** 2009
  *Department of Physics, Humboldt University* Berlin, DE
  Programmed computer simulations of dynamical neuronal networks.

## LANGUAGES

- **Natural** Catalan (native), Spanish (native), English (expert), Italian (fluent), German (basic)
- **Artificial** `R`, `Markdown`, `Quarto`, `LATEX`, `C`, `C++`, `Julia`, `Python`, `PERL`, `HTML`, `PHP`, `Matlab`

## ADDITIONAL TRAINING

- **Oxford Summer School in Economic Networks** 2019
  *Oxford University* Oxford, UK
- **School of Advanced Studies in the Reappraisal of the Surplus Approach** 2018
  *Roma Tre University* Roma, IT

## REFERENCES


- **Willi Semmler**
  Arnhold Professor of International Cooperation and Development
  *New School for Social Research* | New York, NY, US
  semmlerw@newschool.edu | +1-212-229-5717 ext. 3050

- **Marco V. Passarella**
  Associate Professor of Political Economy
  *University of L'Aquila* | L'Aquila, IT
  marco.veronesepassarella@univaq.it | +39 351 714 0730

- **Sakir-Devrim Yilmaz**
  Senior Macroeconomist
  *Agénce Française du Developpément* | Paris, FR
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- **Andrea Genovese**
  Professor of Circular Economy
  *University of Salford Business School* | Salford, Greater Manchester, UK
  a.genovese1@salford.ac.uk | +44 114 222 3347

- **Duncan Foley**
  Leo Model Professor of Economics
  *New School for Social Research* | New York, NY, US
  foleyd@newschool.edu | +1-212-229-5717 ext. 3177